



Assignment 03

Submitted By	Ijlal Ahmad
Submitted To	Ms. Saira Kishwar
Roll No	FL-21565
Date	05-05-2025
Subject	Web Engineering

```
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4      <meta charset="UTF-8">
5      <meta name="viewport" content="width=device-width, initial-scale=1.0">
6      <title>Age & DOB Calculator</title>
7      <style>
8          body {
9              font-family: Arial, sans-serif;
10             max-width: 600px;
11             margin: 0 auto;
12             padding: 20px;
13             background-color: #f5f5f5;
14         }
15         .calculator {
16             background-color: white;
17             padding: 20px;
18             border-radius: 8px;
19             box-shadow: 0 2px 10px rgba(0,0,0,0.1);
20         }
21         h1 {
22             text-align: center;
23             color: #333;
24         }
25         .section {
26             margin-bottom: 30px;
27             padding: 15px;
28             border: 1px solid #ddd;
29             border-radius: 5px;
30         }
31         .section h2 {
32             margin-top: 0;
33             color: #444;
34         }
35         .input-group {
36             margin-bottom: 10px;
```

```

37     }
38     label {
39         display: inline-block;
40         width: 80px;
41         font-weight: bold;
42     }
43     input {
44         padding: 8px;
45         border: 1px solid #ccc;
46         border-radius: 4px;
47         width: 80px;
48     }
49     button {
50         background-color: #4CAF50;
51         color: white;
52         border: none;
53         padding: 10px 15px;
54         text-align: center;
55         text-decoration: none;
56         display: inline-block;
57         font-size: 16px;
58         margin: 10px 0;
59         cursor: pointer;
60         border-radius: 4px;
61         transition: background-color 0.3s;
62     }
63     button:hover {
64         background-color: #45a049;
65     }
66     .result {
67         margin-top: 15px;
68         padding: 10px;
69         background-color: #f8f8f8;
70         border-radius: 4px;
71         border-left: 4px solid #4CAF50;
72     }
73 </style>
74 </head>
75 <body>
76 <div class="calculator">
77     <h1>Age & Date of Birth Calculator</h1>
78
79     <div class="section">
80         <h2>Calculate Age from Date of Birth</h2>
81         <div class="input-group">
82             <label for="dob-day">Day:</label>
83             <input type="number" id="dob-day" min="1" max="31" placeholder="DD">
84
85             <label for="dob-month">Month:</label>
86             <input type="number" id="dob-month" min="1" max="12" placeholder="MM">
87
88             <label for="dob-year">Year:</label>
89             <input type="number" id="dob-year" min="1900" max="2100" placeholder="YYYY">
90         </div>
91         <button id="calculate-age">Calculate Age</button>
92         <div id="age-result" class="result" style="display: none;">
93             Your age is: <span id="age-years">0</span> years, <span id="age-months">0</span> months, and <span id="age-days">0</span> days.
94         </div>
95     </div>
96
97     <div class="section">
98         <h2>Calculate Date of Birth from Age</h2>
99         <div class="input-group">

```

```

99     <div class="input-group">
100         <label for="age-years-input">Years:</label>
101         <input type="number" id="age-years-input" min="0" placeholder="YY">
102
103         <label for="age-months-input">Months:</label>
104         <input type="number" id="age-months-input" min="0" max="11" placeholder="MM">
105
106         <label for="age-days-input">Days:</label>
107         <input type="number" id="age-days-input" min="0" max="30" placeholder="DD">
108     </div>
109     <button id="calculate-dob">Calculate DOB</button>
110     <div id="dob-result" class="result" style="display: none;">
111         Your approximate date of birth is: <span id="dob-date"></span>
112     </div>
113 </div>
114 </div>
115
116 <script>
117     document.getElementById('calculate-age').addEventListener('click', function() {
118         const day = parseInt(document.getElementById('dob-day').value);
119         const month = parseInt(document.getElementById('dob-month').value);
120         const year = parseInt(document.getElementById('dob-year').value);
121
122         if (isNaN(day) || isNaN(month) || isNaN(year)) {
123             alert('Please enter valid day, month, and year');
124             return;
125         }
126
127         const dob = new Date(year, month - 1, day);
128         if (isNaN(dob.getTime())) {
129             alert('Please enter a valid date');

```

```
130         return;
131     }
132
133     const today = new Date();
134     if (dob > today) {
135         alert('Date of birth cannot be in the future');
136         return;
137     }
138
139     let ageYears = today.getFullYear() - dob.getFullYear();
140     let ageMonths = today.getMonth() - dob.getMonth();
141     let ageDays = today.getDate() - dob.getDate();
142
143     if (ageDays < 0) {
144         ageMonths--;
145         // Get the last day of the previous month
146         const lastDayOfMonth = new Date(
147             today.getFullYear(),
148             today.getMonth(),
149             0
150         ).getDate();
151         ageDays += lastDayOfMonth;
152     }
153
154     if (ageMonths < 0) {
155         ageYears--;
156         ageMonths += 12;
157     }
158
159     document.getElementById('age-years').textContent = ageYears;
160     document.getElementById('age-months').textContent = ageMonths;
```

```

161     document.getElementById('age-days').textContent = ageDays;
162     document.getElementById('age-result').style.display = 'block';
163 });
164
165 document.getElementById('calculate-dob').addEventListener('click', function() {
166     const years = parseInt(document.getElementById('age-years-input').value) || 0;
167     const months = parseInt(document.getElementById('age-months-input').value) || 0;
168     const days = parseInt(document.getElementById('age-days-input').value) || 0;
169
170     if (years === 0 && months === 0 && days === 0) {
171         alert('Please enter at least one value (years, months, or days)');
172         return;
173     }
174
175     const today = new Date();
176     let dobYear = today.getFullYear() - years;
177     let dobMonth = today.getMonth() - months;
178     let dobDay = today.getDate() - days;
179
180     if (dobDay < 1) {
181         dobMonth--;
182         // Get the last day of the previous month
183         const lastDayOfMonth = new Date(
184             today.getFullYear(),
185             today.getMonth(),
186             0
187         ).getDate();
188         dobDay += lastDayOfMonth;
189     }
190
191     if (dobMonth < 0) {
192         dobYear--;
193         dobMonth += 12;
194     }
195
196     // Adjust for month being 0-based in JavaScript Date
197     dobMonth++;
198
199     // Format the date
200     const formattedDate = `${dobDay.toString().padStart(2, '0')}/${dobMonth.toString().padStart(2, '0')}/${dobYear}`;
201
202     document.getElementById('dob-date').textContent = formattedDate;
203     document.getElementById('dob-result').style.display = 'block';
204 });
205 </script>
206 </body>
207 </html>

```

Calculate Age from Date of Birth

Day: Month: Year:

Calculate Age

Your age is: 21 years, 10 months, and 19 days.

Calculate Date of Birth from Age

Years: Months: Days:

Calculate DOB

Your approximate date of birth is: 16/06/2003

JavaScript Code Explanation

1. Calculate Age from Date of Birth

- **Purpose:** Calculates a person's age in years, months, and days based on their birth date.
- **How it Works:**
 1. Takes day, month, and year inputs from the user.
 2. Validates the inputs to ensure they form a valid date.
 3. Compares the birth date with the current date to calculate the difference.

4. Adjusts for month and day boundaries (e.g., if the current day is less than the birth day, it borrows days from the previous month).
5. Displays the result in the format: "X years, Y months, and Z days."

2. Calculate Date of Birth from Age

- **Purpose:** Estimates a person's birth date based on their current age in years, months, and days.
- **How it Works:**
 1. Takes years, months, and days inputs from the user.
 2. Validates that at least one value is provided.
 3. Subtracts the age from the current date to estimate the birth date.
 4. Adjusts for month and day boundaries (similar to the age calculator).
 5. Formats the result as "DD/MM/YYYY" and displays it.